

Alexandrene Tanri (she/her/hers)

Vancouver, BC | [LinkedIn.com/in/arenetanri](https://www.linkedin.com/in/arenetanri) | +1 (236) 986 7223 | arenetanri@gmail.com | [Github/Portfolio](#)

EDUCATION

University of British Columbia

Expected Graduation: May 2029

Bachelor of Applied Science – Integrated Engineering

- CGPA: 81.3% (Dean's List).
- Outstanding International Student Award recipient (2025).
- Relevant Courses: Mechanics I (97%), Introductory Physics Laboratory for Engineers (89%), Introduction to Computation in Engineering Design (82%).
- Co-op: Available for 4 months or 12 months beginning May 2027

TECHNICAL SKILLS

Design & Analysis Tools: Solidworks (CAD), Tinkercad, MATLAB, Microsoft Excel

Programming: C, C++, Arduino

Hardware: Microcontrollers, Oscilloscope, Signal Generator, Multimeter, Soldering Iron, Breadboard, Circuit Analysis

Fabrication: 3D Printing, Aviation Snips, Seamer, Nibbler, Hole Punch, Pop Riveter, Hand Tools

TEAM TECHNICAL PROJECTS

Squishy Squeezer – Automated Retrieval Claw | UBC APSC 101

January 2026 - February 2026

- Designed and prototyped an automated retrieval claw in a six-person engineering team, integrating a two-arm gripping mechanism with Arduino-based ultrasonic sensing and servo actuation.
- Fabricated and assembled sheet-metal claw components while iterating arm geometry, servo placement, string attachment, and gripping mechanisms through focused physical prototypes.
- Tested and refined the claw using risk analysis and experimental investigations, including ultrasonic sensor smoothing and adjustable anchor placement to balance gripping width and strength.
- Evaluated competing concepts using engineering requirements, ranking, and a weighted decision matrix to guide selection and development of the Squishy Squeezer.
- Validated the final system in a retrieval competition, increasing the team's score from 3 points in Round 1 to 9 points in Round 2, and documented the design through engineering-drawing review and an e-poster.

RWH System | UBC APSC 101

March 2026 - April 2026

- Designed and modelled a rainwater harvesting system in a six-person team to provide potable water for a remote two-person household in British Columbia.
- Built and analyzed an integrated Excel model incorporating historical rainfall, catchment area, storage capacity, piping, pumps, filtration, disinfection, power requirements, and household water demand.
- Applied fluid-flow and pressure calculations alongside component specifications to evaluate system configurations and quantify interactions between pump performance, piping losses, storage, and water delivery.
- Optimized component selections by balancing cost, reliability, consumption, environmental impact, and stakeholder requirements across the complete system.
- Achieved approximately 80% overall weighted stakeholder satisfaction, including 100% reliability, 94% relative-cost, and 87% consumption satisfaction in the final team design.

Cardboard Chair Project | UBC APSC 100

September 2025 – October 2025

- Designed and fabricated a child-sized cardboard chair in a six-person team for disaster-relief

settings, addressing safety, durability, portability, storage, assembly, and child comfort.

- Generated and prototyped six design concepts and applied quantitative ranking and scoring to select a reinforced cross-base configuration as the final design.
- Developed the final chair with a dual-layer cross-base, backrest supports, rounded edges, storage space, and interlocking features to improve structural support, safety, and ease of assembly.
- Built and evaluated physical prototypes to refine structural geometry and verify the feasibility of producing a functional chair primarily from corrugated cardboard.
- Co-designed the final technical poster with a teammate, communicating stakeholder requirements, concept generation, prototype development, quantitative selection, specifications, and the final design.

Pen Planter | UBC APSC 100

November 2025

- Developed assistive gardening concepts in a six-person team for a wheelchair user with limited hand function, translating user needs into requirements for reach, grip force, weight, stability, and ease of operation.
- Conceptualized an individual seed-dispensing shovel featuring dual handles, extended reach, and a mechanically operated seed chamber before integrating ideas through the team's design process.
- Modelled and prototyped candidate solutions using SolidWorks and physical prototypes to evaluate geometry, usability, manufacturability, and compatibility with the user's mobility constraints.
- Evaluated six concepts using engineering requirements and a weighted decision matrix, supporting selection and refinement of the Pen Planter as the final team design.
- Delivered a lightweight assistive planting concept enabling digging, seed placement, and hole refilling while minimizing the hand force and arm movement required from the user.

VOLUNTEER EXPERIENCE

KR1ZA Voli SMAK 1 Penabur Jakarta | Jakarta, Indonesia

January 2023 - December 2024

Volleyball Team Captain

- Led team practices and coordinated with coaches and teammates to organize training schedules, competition preparation, and team responsibilities.
- Facilitated communication among team members and helped resolve conflicts, fostering a positive and collaborative team environment.
- Led the team to a school cup victory, applying leadership and teamwork throughout interschool competition.

SMAKONECUP | Jakarta, Indonesia

August 2024 - December 2024

Volleyball Cup Committee Coordinator

- Coordinated tournament registration, regulations, and communication with participating schools and coaches to support smooth event preparation.
- Managed event-day logistics and addressed issues with committee members to help maintain tournament operations and scheduling.
- Officiated volleyball matches as a linesperson, applying competition rules and communicating calls with match officials.

INTERESTS & ACTIVITIES

- Have played piano since age 4, achieving Yamaha Grade 7; also play guitar, violin and drums.
- Build and customize mechanical keyboards and PCs, selecting components and assembling systems as a hands-on hobby.
- Build customized mechanical watches, exploring the movement mechanisms and component assembly.